

$$T_{AC} \sin 60 + T_{AB} \sin 45 = 2000$$

$$T_{AC} \cos 60 = T_{AB} \cos 45$$

$$\frac{\sqrt{3}}{2} T_{AC} + \frac{1}{2} T_{AB} = 2000$$

$$T_{AC} \cdot \frac{1}{2} = T_{AB} \cdot \frac{\sqrt{2}}{2}$$

$$\sqrt{3} T_{AC} + T_{AB} = 4000$$

$$T_{AC} = T_{AB} \sqrt{2} \approx 1639,9 \text{ N}$$

$$\sqrt{6} T_{AB} + T_{AB} = 4000$$

$$T_{AB} (\sqrt{6} + 1) = 4000 \rightarrow T_{AB} = \frac{4000 (\sqrt{6} - 1)}{6 - 1} = \frac{4000 \sqrt{6} - 4000}{5} \approx 1159,59 \text{ N}$$

② $A(0,0,10)$ $\vec{AB} = (12, -8, -10) \rightarrow \mu_{AB} = \frac{\vec{r}_{AB}}{|\vec{r}_{AB}|} = \frac{12\hat{i} - 8\hat{j} - 10\hat{k}}{\sqrt{308}} = \frac{12}{17,5}\hat{i} - \frac{8}{17,5}\hat{j} - \frac{10}{17,5}\hat{k} = 0,68\hat{i} - 0,45\hat{j} - 0,57\hat{k}$
 $B(-12, -5, 0)$ $\vec{AC} = (0, 10, -20) \rightarrow \mu_{AC} = \frac{\vec{r}_{AC}}{|\vec{r}_{AC}|} = \frac{0\hat{i} + 10\hat{j} - 20\hat{k}}{\sqrt{200}} = 0\hat{i} + 0,70\hat{j} - 0,70\hat{k}$
 $C(0, 10, 0)$ $\vec{AB} = (12, -5, -10) \rightarrow \mu_{AB} = \frac{\vec{r}_{AB}}{|\vec{r}_{AB}|} = \frac{12\hat{i} - 5\hat{j} - 10\hat{k}}{\sqrt{269}} = 0,45\hat{i} - 0,18\hat{j} - 0,37\hat{k}$
 $D(12, -8, 0)$

$$\sum \epsilon_x: T_{AB}(0,68) + T_{AC}(0) + T_{AD}(-0,37) = 0$$

$$\sum \epsilon_y: T_{AB}(-0,45) + T_{AC}(0,70) + T_{AD}(-0,20) = 0$$

$$2000(0,68) - 0,37T_{AD} = 0 \rightarrow T_{AD} = 1863,9 \text{ N}$$

$$2000(-0,45) + T_{AC} \cdot 0,70 - 1863,9 \cdot 0,20 = 0 \rightarrow T_{AC} = 2089,7 \text{ N}$$

③
 $F_x = F \cos 45 = 700 \cdot \frac{\sqrt{2}}{2} = 495,9 \text{ N}$
 $F_y = F \sin 45 = 700 \cdot \frac{\sqrt{2}}{2} = 495,9 \text{ N}$
 Momento resultante: $M = F \cdot d$
 $\sum \epsilon_x: M_x = F_x \cdot d \rightarrow M_x = 495,9 \cdot 0,5 = 247,95 \text{ N}\cdot\text{m}$
 $\sum \epsilon_y: M_y = F_y \cdot d \rightarrow M_y = 495,9 \cdot 0,5 = 247,95 \text{ N}\cdot\text{m}$
 Momento total: $M = 247,95 + 247,95 = 495,9 \text{ N}\cdot\text{m}$

④ $\vec{AB} = (1, -3, -3) \rightarrow \frac{1\hat{i} - 3\hat{j} - 3\hat{k}}{\sqrt{19}} = 0,23\hat{i} - 0,72\hat{j} - 0,72\hat{k}$
 $\vec{AC} = (1, 3, -3) \rightarrow \frac{1\hat{i} + 3\hat{j} - 3\hat{k}}{\sqrt{19}} = 0,23\hat{i} + 0,72\hat{j} - 0,72\hat{k}$
 $\vec{AD} = (-9, 5, 1) \rightarrow \frac{-9\hat{i} + 5\hat{j} + 1\hat{k}}{\sqrt{91}} = -0,93\hat{i} + 0,53\hat{j} + 0,11\hat{k}$

Equilíbrio de forças:

$$\begin{cases} \sum \epsilon_x: T_{AB} \cdot 0,23 + T_{AC} \cdot 0,23 - T_{AD} \cdot 0,93 = 0 \\ \sum \epsilon_y: -T_{AB} \cdot 0,72 + T_{AC} \cdot 0,72 + T_{AD} \cdot 0,53 = 0 \\ \sum \epsilon_z: -T_{AB} \cdot 0,72 - T_{AC} \cdot 0,72 - T_{AD} \cdot 0,11 + 1000 = 0 \end{cases}$$

* Para o sistema equilibrado

as tensões são iguais

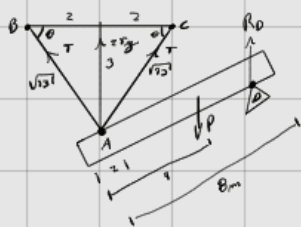
⑤
 $\sum M_A \rightarrow 4000 \cdot 3 - 9 \cdot F_{By} = 0$
 $9 F_{By} = 12000 \rightarrow F_{By} = 1333,3 \text{ N}$
 $F_{Ay} + F_{By} - 4000 = 0$
 $F_{Ay} = 2666,7 \text{ N}$

⑥
 $\sum M_A \rightarrow 20000 \cdot 0,9 + 20000 \cdot 2,8 - 20000 \cdot 3,7 + R_B \cdot 3,7 = 0$
 $R_B = 16666,66 \text{ N}$
 $R_A + R_B - 20000 - 20000 - 20000 = 0$
 $R_A = 74000 - 48666,66 = 25333,34 \text{ N}$

* AVALIAÇÃO 1 - MECÂNICA DOS SÓLIDOS

⑦
 $T_{BC} \cdot \frac{1}{\sqrt{2}} = T_{AC} \cdot \frac{1}{2}$
 $0,707 T_{BC} = 0,5 T_{AC}$
 $T_{AC} = 1,41 T_{BC}$
 $T_{BC} \cdot \frac{1}{\sqrt{2}} + T_{AC} \cdot \frac{1}{2} = 1200$
 $0,707 T_{BC} + 0,707 T_{BC} = 1200$
 $1,414 T_{BC} = 1200 \rightarrow T_{BC} = 849,4 \text{ N}$
 $T_{AC} = 1,41 \cdot 849,4 = 1197,7 \text{ N}$

② Comprimento total: 8m $\rightarrow$ 60 kg/m $\rightarrow$ Massa total = 480 kg $R_D = 4800$



* A tensão em y zero:

$$T_y = 2 \cdot T \cdot \sin \theta$$

$$T_y = 2 \cdot T \cdot \frac{2}{\sqrt{5}} \approx 1,664 T$$

* $\sum M_D = 0$ $\uparrow$

$$1,664 T \cdot 5 - 4800 \cdot 3 = 0$$

$$T \approx 1730,76 \text{ N}$$

* Equilíbrio das forças

$$F_{Ay} + R_D - P = 0$$

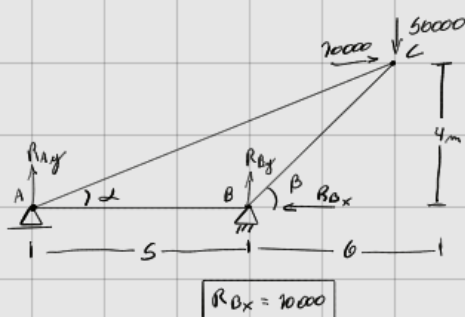
$$1,664 \cdot 1730,76 + R_D - 4800 = 0$$

$$R_D = 1920 \text{ N}$$

③ $R_{AB} = B - A = (0; 1,5; 20) - (0; 0; 0) = (0; 1,5; 20)$

$$M_A = \begin{vmatrix} i & j & k \\ 0 & 1,5 & 20 \\ 3 & 0 & 0 \end{vmatrix} = 30j - 4,5k \rightarrow M_A = \{0i + 30j - 4,5k\} \rightarrow |M_A| = \sqrt{30^2 + (-4,5)^2} = \sqrt{920,25} \approx 30,34 \text{ kN}$$

Lista de exercícios 6:



* $\sum M_B = 0$

$$-5 R_{Ay} - 10000 \cdot 6 - 50000 \cdot 4 = 0$$

$$5 R_{Ay} = -340000$$

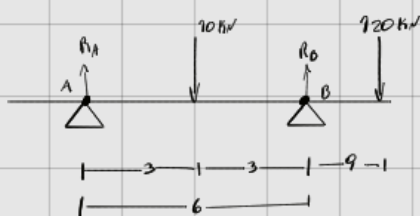
$$R_{Ay} = -68000 \text{ N}$$

* $\sum F_y = 0$

$$R_{Ay} + R_{By} - 50000 = 0$$

$$R_{By} = 50000 + 68000$$

$$R_{By} = 118000 \text{ N}$$



* $\sum M_B = 0$ $\uparrow$

$$-6 R_A + 3 \cdot 10 - 120 \cdot 9 = 0$$

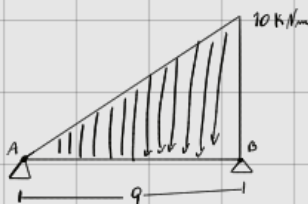
$$6 R_A = 30 - 1080$$

$$R_A = \frac{-1050}{6} = -175 \text{ kN}$$

* $\sum F_y = 0$

$$-175 + R_B - 10 - 120 = 0$$

$$R_B = 305 \text{ kN}$$



$$F_R = \frac{9 \cdot 10}{2} = 45 \text{ kN}$$

Distância em A: $\frac{2}{3} \cdot 9 = 6 \text{ m}$

Distância em B: $\frac{1}{3} \cdot 9 = 3 \text{ m}$

* $\sum M_A = 0$

$$R_B \cdot 9 - F_R \cdot 6 = 0$$

$$9 R_B = 45 \cdot 6$$

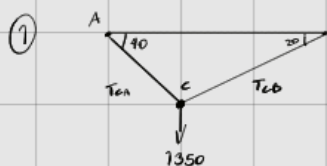
$$R_B = 30 \text{ kN}$$

* $\sum F_y = 0$

$$R_A + R_B - F_R = 0$$

$$R_A = 75 \text{ kN}$$

* Lista 3 mecânica dos sólidos:



$$T_{CB} \cdot \cos 20 = T_{CA} \cdot \cos 40 \rightarrow T_{CB} = T_{CA} \cdot \frac{\cos 40}{\cos 20} = 0,81 T_{CA}$$

$$T_{CB} \cdot \sin 20 + T_{CA} \cdot \sin 40 = 1350$$

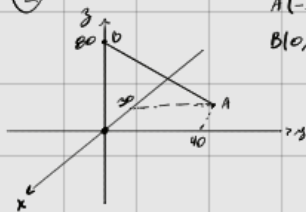
$$T_{CB} = 1207,23 \text{ N}$$

$$0,81 T_{CA} \cdot \sin 20 + \sin 40 T_{CA} = 1350$$

$$0,91 T_{CA} = 1350$$

$$T_{CA} = 1483 \text{ N}$$

②



$$A(-30, 40, 0) \quad r_{AB} = (0, 0, 80) - (-30, 40, 0)$$

$$B(0, 0, 80) \quad r_{AB} = (30, -40, 80)$$

$$\mu_{AB} = \frac{30\hat{i} - 40\hat{j} + 80\hat{k}}{94,2} = 0,31\hat{i} - 0,42\hat{j} + 0,84\hat{k}$$

$$F_x = 0,31 \cdot 2500 = 775$$

$$F_y = -0,42 \cdot 2500 = -1050$$

$$F_z = 0,84 \cdot 2500 = 2100$$

$$\theta_x = \arccos \frac{775}{2500}$$

$$\theta_y = \arccos \frac{-1050}{2500}$$

$$\theta_z = \arccos \frac{2100}{2500}$$



③

$$r_{AB} = B - A = (0, 8; 0; 1,6) - (3,6; 1,6; 0) = (-3,6; -1,6; 1,6) \rightarrow \mu_{AB} = \frac{-3,6\hat{i} - 1,6\hat{j} + 1,6\hat{k}}{3,6}$$

$$F_{AB} = (-9,77\hat{i} - 4,44\hat{j} + 4,44\hat{k}) \cdot 900$$

$$F_{ABx} = -8800 \text{ N}; F_{ABy} = -4000 \text{ N}; F_{ABz} = 4000 \text{ N}$$

$$r_{CB} = (0, 8; 0; 1,6) - (0; 1,6; 0) = (0,8; -1,6; 1,6) \rightarrow \mu_{CB} = \frac{0,8\hat{i} - 1,6\hat{j} + 1,6\hat{k}}{2,4}$$

$$F_{CB} = (0,33\hat{i} - 0,66\hat{j} + 0,66\hat{k}) \cdot 150$$

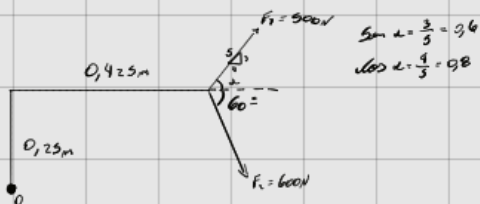
$$F_{CBx} = 50 \text{ N}; F_{CBy} = -100 \text{ N}; F_{CBz} = 100 \text{ N}$$

④

$$r_{AB} = (0, 2; 10) - (0, 0; 0) = (0, 2; 10)$$

$$M = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 2 & 10 \\ 200 & 0 & 0 \end{vmatrix} = 2000\hat{j} - 400\hat{k} \rightarrow |M| = \sqrt{(2000)^2 + (-400)^2} \approx 2039,6 \text{ N}$$

⑤



* F_{1x} :

$$F_{1x} = 900 \cdot 0,8 = 720 \text{ N}$$

$$F_{1y} = 900 \cdot 0,6 = 540 \text{ N}$$

* F_{2x} :

$$F_{2x} = 500 \cdot 0,8 = 400 \text{ N}$$

$$F_{2y} = 500 \cdot 0,6 = 300 \text{ N}$$

$$\sum M_0 = 0 \rightarrow \left. \begin{aligned} & 900 \cdot 0,425 = 382,5 \\ & -900 \cdot 0,25 = -225 \end{aligned} \right\} 2,5 \text{ m} \quad \left. \begin{aligned} & -579,62 \cdot 0,425 = -246,34 \text{ N} \cdot \text{m} \\ & -300 \cdot 0,25 = -75 \text{ N} \end{aligned} \right\} -295,89 \text{ N} \cdot \text{m}$$

$$M_0 = 382,5 - 295,89 = 86,61 \text{ N} \cdot \text{m}$$